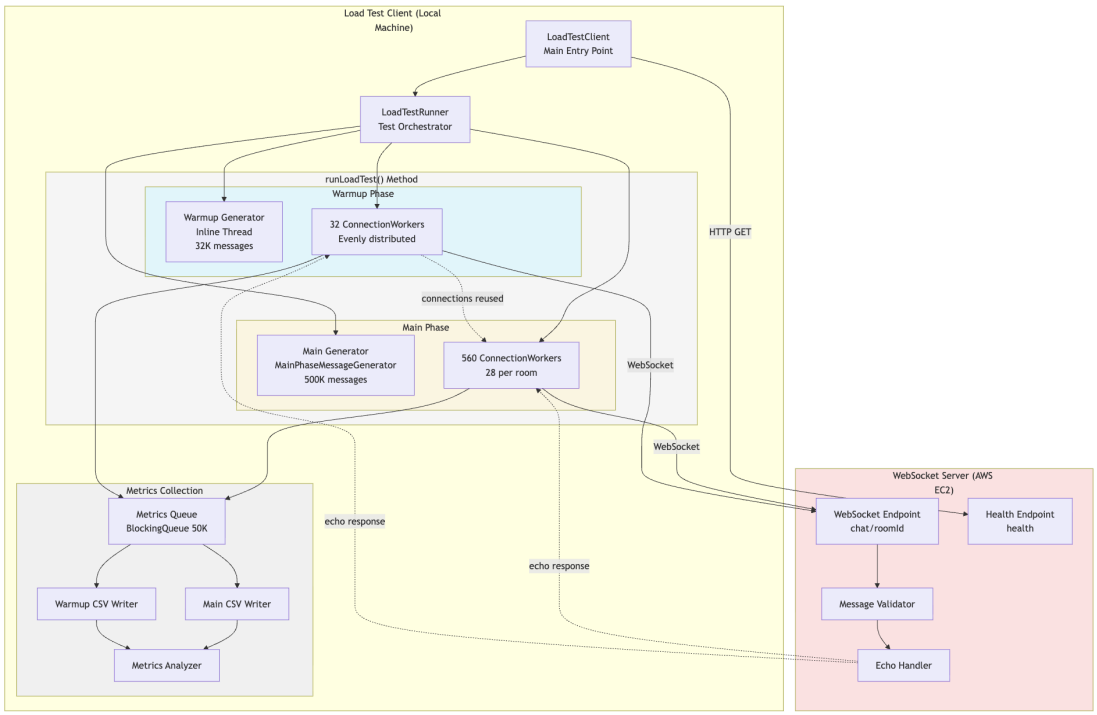
Git Repository URL:

https://github.com/LelinZheng/ChatFlow_CS6650

Design Document:

Architecture Overview:

The ChatFlow load test system is a multithreaded WebSocket client that simulates high-volume concurrent chat traffic across 20 rooms. The system executes two phases within a single method: warmup establishes 32 persistent connections distributed evenly across rooms, then the main phase adds 528 connections to reach 28 per room (560 total), processing 500,000 messages with random room distribution.



Major classes and their relationships:

LoadTestClient: Entry point that performs HTTP health check and initiates testing.

LoadTestRunner: Orchestrates test execution via `runLoadTest()`. Creates thread pool, manages both test phases, coordinates message generation, and aggregates statistics. Implements connection reuse by keeping warmup workers alive.

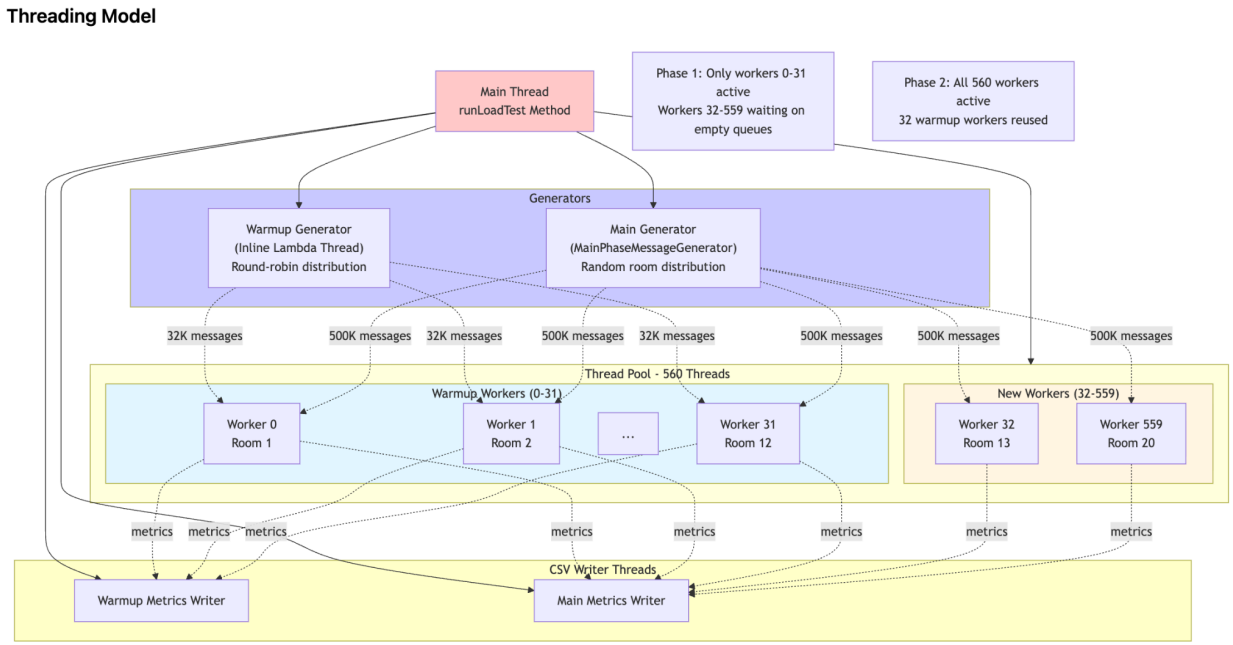
ConnectionWorker: Each owns one persistent WebSocket connection. Uses send-then-wait-for-echo protocol with single-slot mailbox for latency measurement. Implements exponential backoff retry for both initial connection and message sends. Failed workers drain their queues and count messages as failures.

MainPhaseMessageGenerator: Dedicated thread producing 500K messages with random room assignment, routing to workers via per-room round-robin distribution.

MessageFactory: Generates randomized messages (userId 1-100K, 50-message pool, 90% TEXT/5% JOIN/5% LEAVE distribution).

MetricRecord & CsvMetricsWriter: Capture per-message latency and write to CSV asynchronously via blocking queue.

MetricsAnalyzer: Computes statistics (mean, median, p95, p99 latency, per-room throughput, time-series buckets).



Thread Pool: Fixed size (560 threads), created once, shared across phases.

Main Thread: Executes `runLoadTest()`, coordinates phases, waits for queue draining via polling.

Generator Threads (2 sequential):

- Warmup: Inline lambda, produces 32K messages, round-robin to workers 0-31 (1,000 each)
- Main: MainPhaseMessageGenerator, produces 500K messages, random rooms

Worker Threads (560):

- Warmup: Workers 0-31 active, 32-559 blocked on empty queues
- Main: All 560 active processing messages

- Protocol: Block on `queue.take()` when idle, enabling reuse without CPU consumption

Metrics Threads (2 sequential): Warmup and main CSV writers consume from shared metrics queue.

WebSocket connection management strategy:

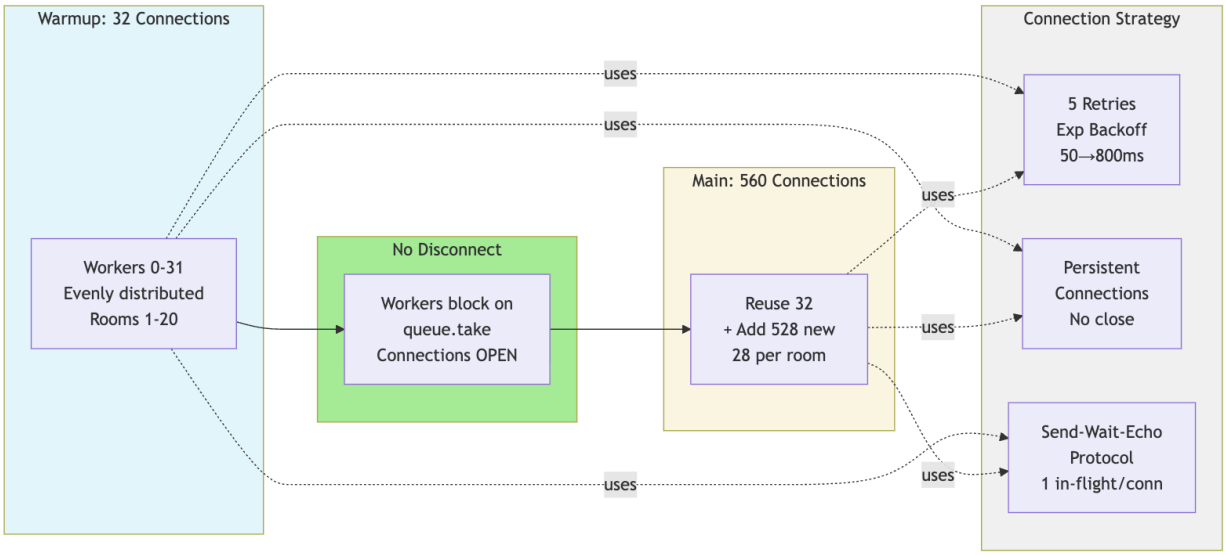
Warmup Distribution: 32 workers assigned via `i % 20 + 1`, yielding rooms 1-12 with 2 workers, rooms 13-20 with 1 worker.

Connection Persistence: After warmup, workers block on `queue.take()` rather than closing connections. This avoids TCP TIME_WAIT state (30-120s) and prevents port exhaustion.

Main Phase Fill: Round-robin fill algorithm iterates rooms 1-20 repeatedly, adding workers until each reaches `connPerRoom=28`. Result: exactly 28 workers per room.

Error Handling: 5 retries with exponential backoff for connections and sends. Dead workers drain queues and count messages as failures, ensuring `OK + failed = total messages`.

Protocol: Send-then-wait-for-echo with 5s timeout ensures sequential processing and accurate latency measurement..



Little's Law calculations and predictions

Little's Law: L = λ × W, where L = concurrent connections, λ = throughput (msg/s), W = average time per message. The Measured RTT (W) is 33.65ms (from main phase mean latency) and the Concurrent connections (L) is 560. Theoretical Maximum: λ = L / W = 560 / 0.03365s = 16641.90 msg/s

The measured throughput is 14728.54 msg/s, which is 88.5% of the theoretical value. The 12% gap from theoretical maximum results from: (1) random room distribution causing occasional queue contention, (2) metrics collection overhead, and (3) network jitter (15-293ms range).

Test Results:

- Screenshot of Part 1 output (basic metrics)

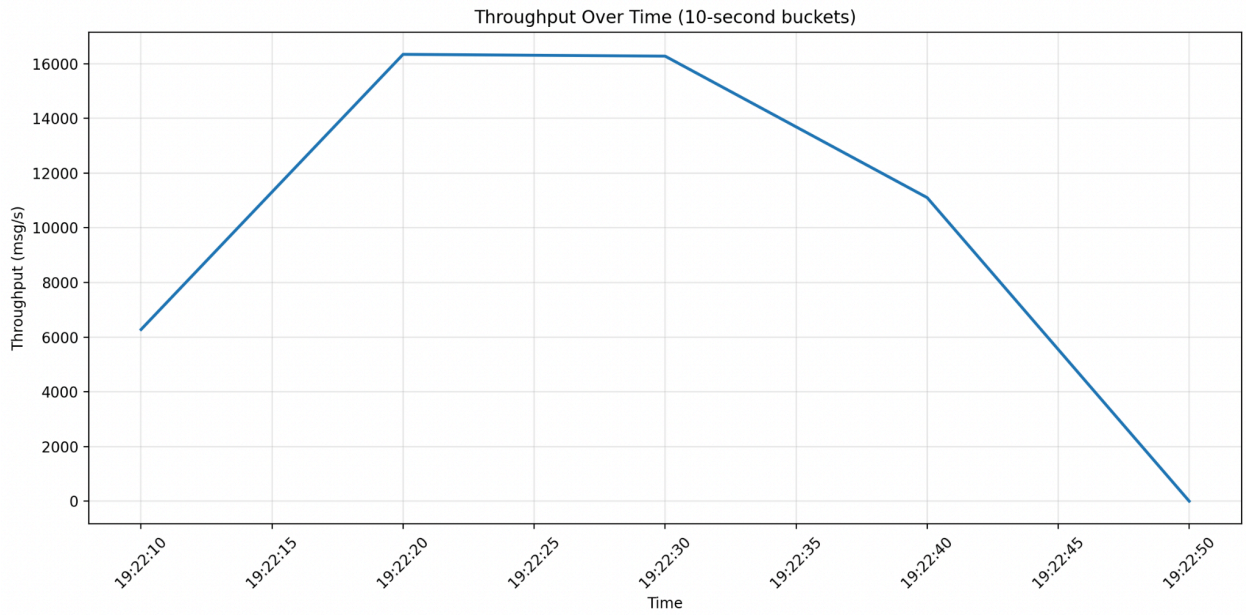
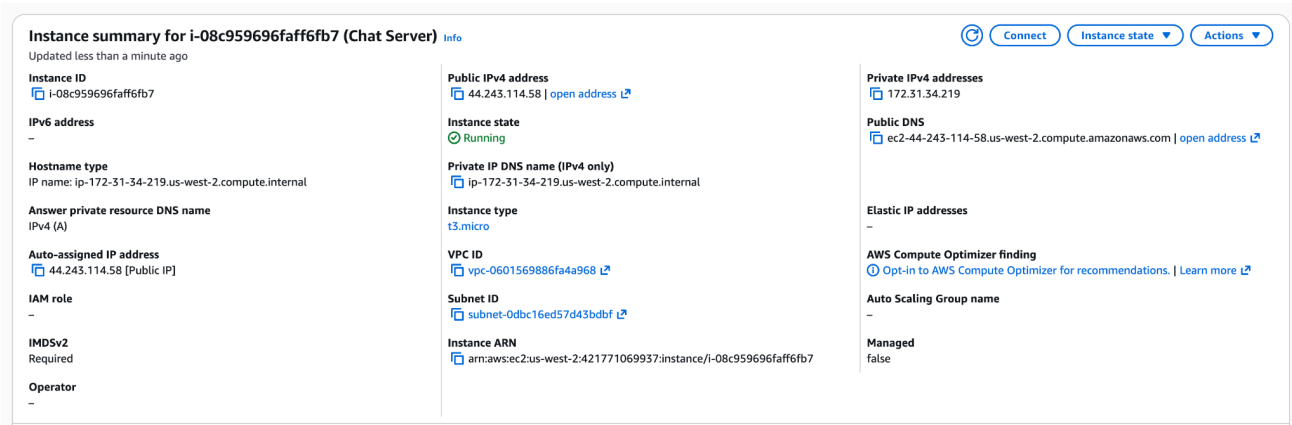
```
Load Test Client Started
Performing server health check...
✓ Server health check passed: {"status":"UP","timestamp":"2026-02-10T19:09:53.751799630Z"}
Starting load test...

=== Starting Warmup Phase ===

=== Warmup Phase Complete ===
OK=32000 failed=0
timeSec=30.77
throughput msg/s=1039.84
Warmup connections remain open for reuse.

=== Starting Main Phase ===

All workers ready. Generating main phase messages and sending them...
```

```
=== Main Phase Results ===
OK=500000 failed=0
timeSec=33.95
throughput msg/s=14728.54
connections=560
reconnections=0
connectionFailures=45
deadWorkers=0
warmupConnectionsReused=32

Process finished with exit code 0
```

- Screenshot of Part 2 output (detailed metrics)

```
Load Test Client Started
Performing server health check...
✓ Server health check passed: {"status":"UP","timestamp":"2026-02-10T19:21:44.503562862Z"}
Starting load test...

=== Starting Warmup Phase ===

=== Warmup Phase Complete ===
OK=32000 failed=0
timeSec=30.70
throughput msg/s=1042.29
Warmup connections remain open for reuse.

=== Starting Main Phase ===

All workers ready. Generating main phase messages and sending them...

=== Main Phase Results ===
OK=500000 failed=0
timeSec=35.30
throughput msg/s=14164.54
connections=560
reconnections=0
connectionFailures=0
deadWorkers=0
warmupConnectionsReused=32
Load test summary:
Main phase metrics to be aggregated here.
=== Metrics Summary ===
OK=500000 failed=0
durationSec=35.187
=== Latency (OK only, ms) ===
mean=33.655
median=33.000
p95=41.000
p99=51.000
min=16.000 max=213.000
=== Throughput per room (OK only, msg/s) ===
room 1: 710.97
room 2: 701.79
room 3: 710.75
room 4: 713.39
room 5: 714.33
room 6: 714.55
room 7: 713.53
room 8: 703.70
room 9: 707.53
room 10: 713.59
room 11: 716.94
room 12: 713.45
room 13: 706.60
room 14: 713.90
room 15: 702.56
room 16: 711.31
room 17: 715.29
room 18: 714.18
room 19: 704.04
room 20: 707.39
=== Message type distribution (all statuses) ===
JOIN: 24947
TEXT: 449786
LEAVE: 25267
Throughput buckets CSV: /Users/lelinzheng/Desktop/Northeastern/Distributed_CS6650/ChatFlow_CS66
50/client-part2/../results/throughput_10s.csv
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

- Performance analysis charts:



- Evidence of EC2 deployment